

REGRESSION ANALYSIS (LECTURE 4)



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Simple Linear Regression Line Using Method of Least Squares:

We assume the equation of Regression Line of Y on X as $y = a + bx$ and the equation of the Regression Line of X on Y as $x = c + dy$. Let us assume that we are given with a bivariate data $(x_i, y_i); i = 1, 2, 3, \dots, N$ for the variables X and Y . We have to estimate the parameters a, b, c and d using the given bivariate data applying the Method of Least Squares. Applying the Method of Least Squares for the Regression Line of Y on X we get the normal equations as:

$$\sum_{i=1}^N y_i = Na + b \sum_{i=1}^N x_i$$

$$\sum_{i=1}^N x_i y_i = a \sum_{i=1}^N x_i + b \sum_{i=1}^N x_i^2$$

Solving the above system we can get the corresponding values of a and b .

Again applying the Method of Least Squares for the Regression Line of X on Y we get the normal equations as:

$$\sum_{i=1}^N x_i = Nc + d \sum_{i=1}^N y_i$$
$$\sum_{i=1}^N x_i y_i = c \sum_{i=1}^N y_i + d \sum_{i=1}^N y_i^2$$

Solving the above system we can get the corresponding values of c and d .

Problem: Obtain both the regression equations for the given bivariate data using Method of Least Squares:

X: 3 2 7 5 8

Y: 6 1 8 6 9

Also estimate y for $x = 11$ and x for $y = 10$.

Answer: Here we have to find out the two regression lines Y on X and X on Y . To estimate y for $x = 11$ we will use regression line of Y on X and to estimate x for $y = 10$ we will use regression line of X on Y .

Given bivariate data :

X	3	2	7	5	8
Y	6	1	8	6	9

We frame the following table :

X	Y	X^2	Y^2	XY
3	6	9	36	18
2	1	4	1	2
7	8	49	64	56
5	6	25	36	30
8	9	64	81	72
$\Sigma X = 25$	$\Sigma Y = 30$	$\Sigma X^2 = 151$	$\Sigma Y^2 = 218$	$\Sigma XY = 178$

So for the regression line of Y on X the normal equations are

$$\Sigma Y = Na + b \Sigma X$$

$$\Sigma XY = a \Sigma X + b \Sigma X^2$$

$$\text{i.e. } 30 = 5a + 25b \quad [\text{Here } N=5]$$

$$178 = 25a + 151b$$

$$\text{Solving we get } a = \frac{8}{13} \text{ and } b = \frac{14}{13}$$

So the equation of the regression line of Y on X is

$$Y = a + bx \quad \text{i.e.} \quad Y = \frac{8}{13} + \frac{14}{13}x$$

$$\Rightarrow 13Y = 8 + 14x \quad \text{or,} \quad \boxed{14x - 13Y + 8 = 0}$$

Again for the regression line of X on Y the normal equations are

$$\Sigma x = Nc + d \Sigma y$$

$$\Sigma xy = \Sigma y^2 + d \Sigma y^2$$

$$\text{i.e.} \quad 25 = 5c + 30d \quad [\text{Here } N=5]$$

$$178 = 30c + 218d$$

$$\text{Solving we get } c = \frac{11}{19} \text{ and } d = \frac{14}{19}$$

So the equation of the regression line of X on Y is

$$x = c + dy \quad \text{i.e.} \quad x = \frac{11}{19} + \frac{14}{19}y$$

$$\Rightarrow 19x = 11 + 14y \quad \text{or,} \quad \boxed{19x - 14y = 11}$$

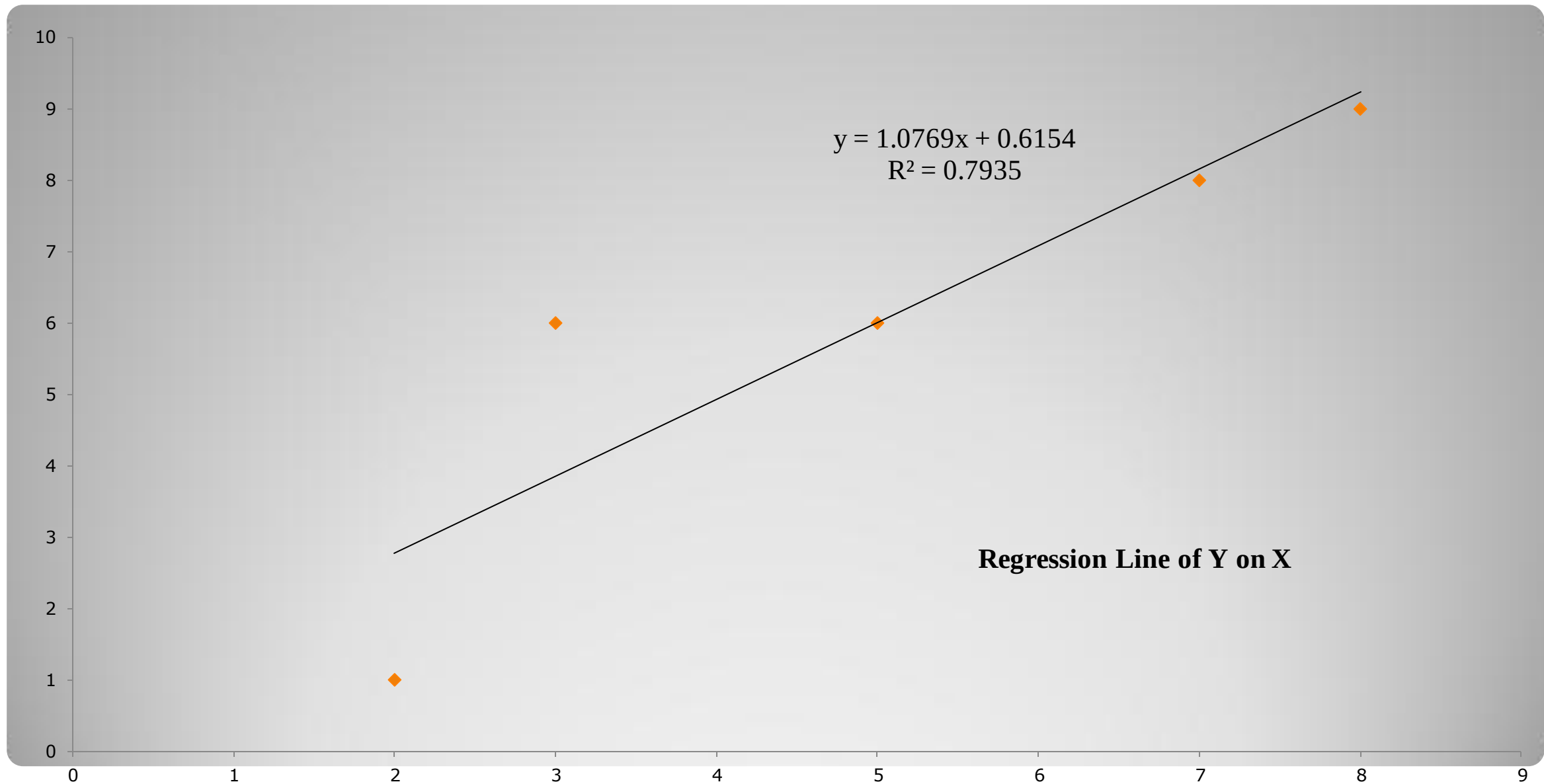
To estimate y for $x=11$ we use regression line of Y on X .

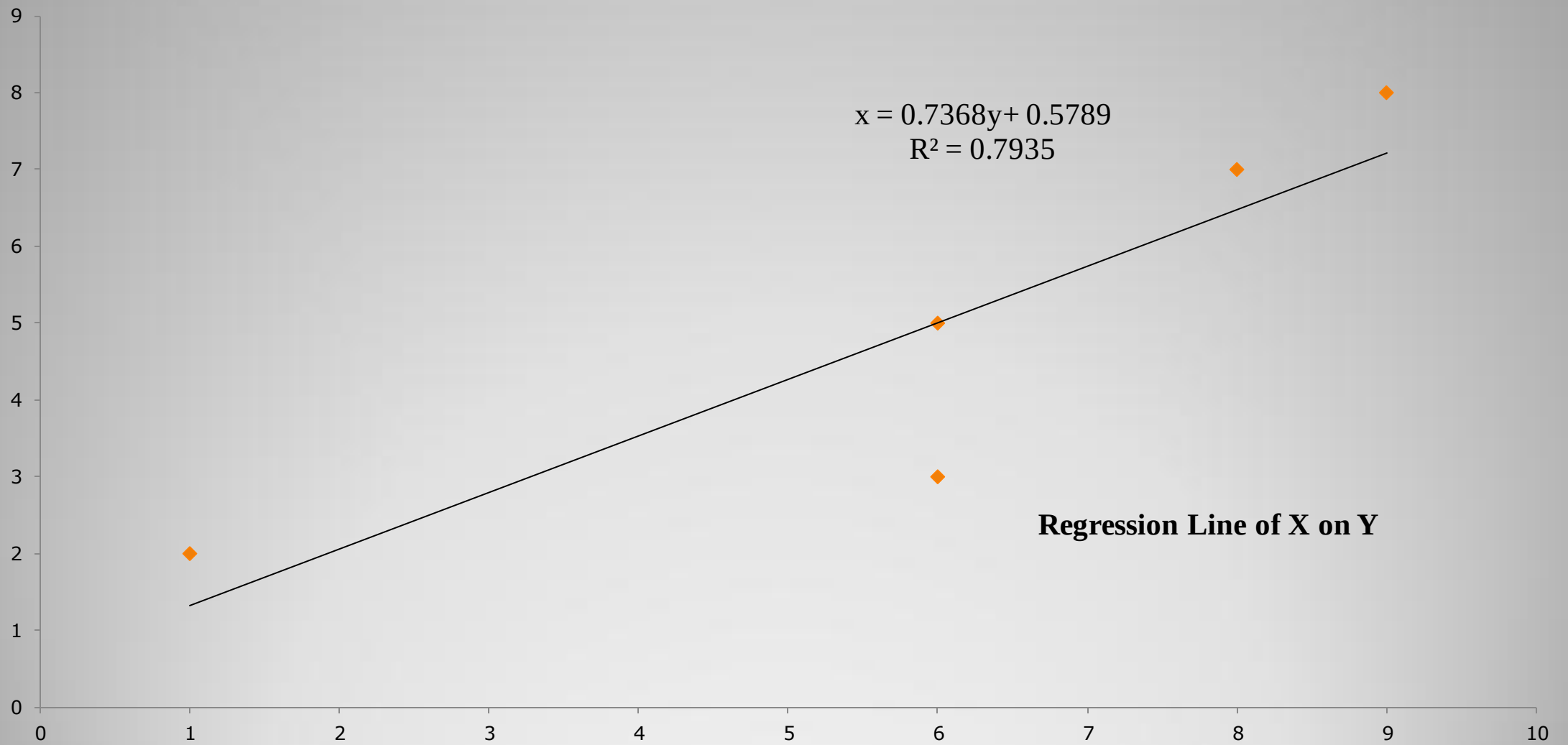
$$\text{So we get, } 14 \times 11 - 13y + 8 = 0 \Rightarrow 13y = 162$$

$$\Rightarrow y = \frac{162}{13} = 12.46$$

To estimate x for $y=10$ we use regression line of X on Y

$$\text{So we get, } 19x - 14 \times 10 = 11 \Rightarrow 19x = 151 \Rightarrow x = \frac{151}{19} = 7.95$$





Derivation of Normal Equations Using Least Square Method for Regression line of Y on X:

We consider the equation of the regression line of Y on X as $Y = a + bX$. We consider the bivariate data as

$(X_i, Y_i): i=1, 2, 3, \dots, N$. For $x = x_i$, we consider the estimated y using the regression line as $y_i^* = a + bx_i$, $i=1, 2, 3, \dots, N$. Now, the deviation between estimated y and actual y will be $y_i^* - y_i$; $i=1, 2, \dots, N$.

So, average (mean) squared deviation

$$= \frac{1}{N} \sum_{i=1}^N (y_i^* - y_i)^2 = \frac{1}{N} \sum_{i=1}^N (a + bx_i - y_i)^2 = S \text{ (say)}.$$

We want to make the mean squared deviation minimum. For that we must have

$$\frac{\partial S}{\partial a} = 0 \Rightarrow \frac{2}{N} \sum_{i=1}^N (a + bx_i - y_i) = 0 \Rightarrow \sum_{i=1}^N (a + bx_i - y_i) = 0$$
$$\Rightarrow Na + b \sum_{i=1}^N x_i - \sum_{i=1}^N y_i = 0 \Rightarrow \boxed{\sum_{i=1}^N y_i = Na + b \sum_{i=1}^N x_i} \quad (1)$$

$$\text{Again, } \frac{\partial S}{\partial b} = 0 \Rightarrow \frac{2}{N} \sum_{i=1}^N x_i (a + bx_i - y_i) = 0$$
$$\Rightarrow \sum_{i=1}^N (ax_i + bx_i^2 - x_i y_i) = 0 \Rightarrow \boxed{\sum_{i=1}^N x_i y_i = a \sum_{i=1}^N x_i + b \sum_{i=1}^N x_i^2} \quad (2)$$

Similarly we can derive the normal equations using Least Square Method for Regression line of X on Y.